Masking Weak Identification: Spectral Gap Lifting from an Omitted Variable

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Abstract

Identification through changes in volatility rests on the separation of the generalised eigenvalues – the *volatility ratios* – of a covariance pencil; in the simple bivariate design of Lewis (2022) the weak-identification diagnostic is a monotone function of the eigenvalue *gap*. Here, we study how a single omitted variable perturbs this gap: our concern is to see how leaving out a relevant variable in the DGP, a classic problem in statistical and econometric analysis, can affect pre-tests of identification's strength. We first present the problem in the same context of Lewis (2022), then proceed with a first-order analysis and an exact secular equation. In the weak identification regime, the result is clean: at exact degeneracy the omitted variable lifts the gap to a strictly positive value whenever it is heteroskedastic relative to the shocks and not orthogonal to the degenerate subspace. This is the precise sense in which misspecification can mask weak identification. A Monte Carlo experiment built on the replication code of Lütkepohl et al. (2021) shows that the distortion is a size distortion that worsens with the sample size, and that a single omitted variable can render a wholly unidentified system empirically indistinguishable from a partially identified one.

1 Introduction

Identification of structural shocks by exploiting changes in their second moments begins with Sentana and Fiorentini (2001) for conditionally heteroskedastic factor models and with Rigobon (2003) for SVAR models. Rigobon and Sack (2004) provide the canonical application to monetary policy and asset prices. The device is now standard in the empirical literature (Kilian and Lütkepohl, 2017): if the covariance matrices of the VAR reduced-form innovations differ across two volatility regimes, $\Sigma_1 = BB'$ and $\Sigma_2 = B\Lambda B', \Lambda = \text{diag}(\lambda_1, ..., \lambda_n)$, then the structural impact matrix B is identified up to column order and sign change whenever the volatility ratios λ_i are pairwise distinct. Equivalently, and as used throughout this paper, the λ_i are the generalised eigenvalues of the covariance pencil (Σ_2, Σ_1) and the columns of B its generalised eigenvectors.

Because identification relies on distinctness of the λ_i, and because the data are in principle informative about it, the condition is testable. Lanne and Lütkepohl (2008) assess this informally, by inspecting whether the estimated volatility ratios have non-overlapping standard-errors intervals. Lütkepohl et al. (2021) supply the formal counterpart: a Wald-type test built on the ratio of the geometric to the arithmetic mean of the estimated eigenvalues. Two features of their test matter. First, its null hypothesis is *lack* of identification, i.e., perfect equality in the generalised eigenvalues, so rejection is what supports identification. Second – the point of contact with the present work – the statistic is a function of the observed covariance matrices $\widehat{\Sigma}_1, \widehat{\Sigma}_2$,

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and correct specification of the two-regime decomposition is maintained throughout their paper. Near-violations of distinctness equally disrupt inference. Notably, Lewis (2022) formalises weak identification in this setting – variances that change very little, or several variances changing nearly proportionally – derives the resulting non-standard behaviour of the estimator, and proposes a diagnostic for it[1].

In this paper, we study what happens to diagnostics of identification strength when the econometrician misspecifies the DGP of interest by leaving out an economically relevant variable. Importantly, these consequences are indeed related to, yet distinctly disruptive compared to, the standard bias introduced in the estimates of the VAR's coefficients. We ask whether it is possible to generate spurious identifying power from a completely degenerate (i.e., non-identified) DGP due to such an omission. We find that the contamination of the residuals caused by a heteroskedastic omitted variable systematically leads to the rejection of the true null hypothesis of non-identification. This happens because current pre-tests evaluate the observed, perturbed covariances. Consequently, a successful 'pass' from these diagnostics does not automatically validate the model; the apparent identification might stem not from the DGP of interest, but from a wholly unrelated channel of contamination. Therefore, researchers must carefully investigate, by both statistical and theoretical tools, the need for additional controls before relying on tested, strong spectral separation.

To formally establish this masking phenomenon, we follow different methodologies. First, we model the omitted variable as a rank-one additive perturbation to the covariance matrices of a structural VAR. This simplification keeps the algebra at a tractable level but implies that lagged dynamics are not accounted for. We address this extension in Appendix B. After setup, we begin our discussion presenting the issue of masking in the framework of Lewis (2022), which recasts heteroskedastic identification as a problem of instrumental variables (IV). The advantage of the IV recast is that it is a very simple illustration where the consequences of omission are immediately intuitive and at the same time difficult to ignore. Then, through a first-order analysis relying on the Courant-Fischer minimax principle and the envelope theorem, we highlight two key features that account for a disruptive perturbation: geometric alignment and heteroskedasticity. Finally, we derive an exact secular equation to study the generalised eigenvalues of the perturbed matrix pencil. Our core theoretical result proves that, at exact degeneracy, the omitted variable lifts the spectral gap to a strictly positive value whenever it is heteroskedastic relative to the structural shocks and not orthogonal to the degenerate subspace. To take sampling noise and uncertainty of empirical estimates into account, a Monte Carlo experiment building on the replication code of Lütkepohl et al. (2021) shows that this mechanism induces a severe size distortion in standard tests, worsening with the sample size. Importantly, we observe that a single omitted variable can render a wholly unidentified system empirically indistinguishable from a partially identified one.

The remainder of the paper is thus organised as follows. Section 3 sets up the notation and the omitted variable perturbation framework. Section 4 recasts the problem in the bivariate instrumental variable setting of Lewis (2022) to provide intuition on the omitted variable wedge. Section 5 develops the interpretable first-order analysis of the Rayleigh quotient. Section 6 provides the rigorous derivation of the exact secular equation and establishes the main degeneracy lifting theorem. Section 7 presents the Monte Carlo evidence. Section 8 concludes.

2 Background literature

Work on misspecification of this identification scheme is scarce, and what exists generally concerns objects other than an omitted variable. Sims (2020) studies misspecification in the individuation

[1] Lewis (2025) offers an up-to-date review of existing diagnostics of weak and non-identification.

of structural breaks between volatility regimes, and finds that such mistakes *compress* the dispersion of the estimated volatility ratios, which underlines the non-obvious nature of the present discussion. It is not clear a priori that misspecification should widen rather than narrow the spread of eigenvalues. Indeed, we will see that this result is only valid for near-degenerate settings – which nevertheless happen to be the only dangerous ones in terms of identification strength. Bacchiocchi et al. (2024) discuss identification behaviour under eigenvalue multiplicity – the case in which only a subset of shocks exhibits a variance break – but again without an omitted variable perturbation.

That omitting variables from a VAR as well as from any other statistical model is consequential is of course well understood; the channel usually emphasised is invertibility. When the innovations span a smaller space than the structural shocks, it is not possible to recover the latter precisely, and the representation is non-fundamental (Lippi and Reichlin (1994), Fernández-Villaverde et al. (2007), Sims (2012) and Forni and Gambetti (2014)). This is distinct from the channel studied here, which operates through the covariance pencil's spectrum. In any case, this acknowledgement shows that the omission of a volatile, correlated variable is a routine occurrence in applied SVAR work rather than a niche scenario.

The general phenomenon that a misspecified model can inflate a measure of identification strength, and so return a false pass, is documented outside the SVAR literature. In cross-sectional asset pricing, a useless or omitted factor can produce spuriously large R^2 and t-statistics that mimic strong identification: see Kan and Zhang (1999), Kleibergen and Zhan (2020), and Kleibergen (2009).

Our paper is naturally motivated by the broad literature on inference under weak identification – see e.g. Lewis (2022) for heteroskedastic identification and Chevillon, Mavroeidis, and Zhan (2020) for long-run restrictions – and weak instruments. This latter branch of the literature has been particularly fertile in recent years, with many relevant and much discussed contributions. Staiger and Stock (1997) and Stock and Yogo (2002) show that when identifying information is scarce, conventional OLS/IV asymptotics become non-standard and confidence intervals become severely distorted. For further recent developments in the field of study, see Andrews, Stock, and Sun (2019) or Lewis and Mertens (2026).

2.1 Positioning

Against this background, the contribution of this paper can be assessed. The literature has established that identification via heteroskedasticity requires distinct volatility ratios, how to test that requirement formally (Lütkepohl et al., 2021) and how to conduct inference when the requirement is nearly violated (Lewis, 2022). What is missing is an analysis of how the identifying object itself, the spectral gap, responds to misspecification of the set of variables. This paper fills that gap for the leading case of a single omitted variable: at exact degeneracy, a heteroskedastic omitted variable does not leave the spectrum alone nor does it compress it, but *lifts* the gap to a strictly positive value, provided that the omitted shock is heteroskedastic relative to the other shocks and not orthogonal to the degenerate subspace. Because the diagnostics are computed on the observed covariances, the lifted gap is what they see. The results are therefore complementary to, not in competition with, that work: those papers provide the correct inference on the identifying information under the hypothesis of correct specification, while the mechanism described here concerns what that information measures when the variable set is incomplete. In the language of the Wald test of Lütkepohl et al. (2021), the consequence is a size distortion – spurious rejection of a true null of nonidentification.

3 Setup and notation

Let the observed data be a vector of serially uncorrelated, mean-zero innovations η_t. These could be observed or as-if observed (from a consistently estimated VAR). In the two volatility regimes, the covariance matrices are Σ_1, Σ_2 respectively, both symmetric positive definite (SPD). Identification through Changes in Volatility (CV) in the Rigobon (2003) framework is simply the generalised eigenvalue problem (GEP)

$$\Sigma_2 q_i = \lambda_i \Sigma_1 q_i, \qquad 0 < \lambda_1 \leq \cdots \leq \lambda_n. \tag{1}$$

We normalise the generalised eigenvectors in the Σ_1 metric. The λ_i are the regime-2/regime-1 variance ratios of the structural shocks; the strength of identification is governed by their separation, measured by the gap

$$\delta_{ij} = |\lambda_i - \lambda_j|, \qquad \delta_j = \min_{i \neq j} |\lambda_i - \lambda_j|. \tag{2}$$

In a bivariate case, we write simply $\delta = \lambda_2 - \lambda_1 > 0$. A geometric reading of (1)–(2) will be useful later. CV identification consists of two steps: a whitening performed through a Cholesky factor and a spectral decomposition. Whitening maps the regime-1 covariance to the identity, so that the one-standard-deviation surface of the regime-2 covariance is an ellipsoid whose principal axes correspond to the structural directions q_i and whose semi-axes have lengths $\sqrt{\lambda_i}$. The gap (2) is then a measure of the local eccentricity of that ellipsoid: identification fails precisely when two semi-axes coincide, so that the corresponding section degenerates into a circle within which no pair of orthogonal axes is distinguishable from any other.

We now proceed to model our omission framework.

Assumption 1 (Contemporaneous omission)**.** The true DGP for the full $(n+1)$ variable system $(y_t', f_t)'$ is a stationary VAR(p), $p \geq 0$, with mutually orthogonal structural shocks and a time-invariant contemporaneous impact matrix. The omitted variable f_t enters the equations of the n observed variables y_t only contemporaneously: the lag polynomial linking $f_{t-1}, \ldots, f_{t-p}$ to y_t is identically zero.

Assumption 1 restricts the channel through which f_t affects y_t, not the dynamics of y_t itself, which can remain a general VAR(p) for arbitrary p. Because f_t never enters through the lags, the VAR(p) estimated on the n observed variables alone is correctly specified in its dynamic part: the reduced-form coefficients are consistently estimated, $\hat{A}_i \xrightarrow{p} \tilde{A}_i$ for $i = 1, \ldots, p$, and the resulting residuals are, asymptotically, serially uncorrelated. The only surviving channel of contamination is therefore contemporaneous. The lag order p plays no further role.

The omitted-variable perturbation. As a consequence of the assumption above, which rules out lagged dynamics, we can write a SVAR(0)

$$\eta_t \;=\; \begin{pmatrix} \eta_{1t} \\ \vdots \\ \eta_{nt} \\ f_t \end{pmatrix} \;=\; \begin{bmatrix} B_{11} & \cdots & B_{1n} \\ \vdots & \ddots & \vdots \\ B_{n1} & \cdots & B_{nn} \end{bmatrix} \begin{pmatrix} \varepsilon_{1t} \\ \vdots \\ \varepsilon_{nt} \end{pmatrix} \;+\; b\, f_t \;=\; B\varepsilon_t + b f_t,$$

Integrating out one omitted variable adds a covariance component to each regime, so instead of (Σ_1, Σ_2) the econometrician observes

$$\widehat{\Sigma}_r = \Sigma_r + R_r, \qquad r = 1, 2. \tag{3}$$

With a single omitted variable the injection is rank-one along a *fixed* (as stated in Assumption 1, we adopt the hypothesis of time-invariance of the structural impact matrix, which is common in the SVAR literature) loading direction b, with only its volatility changing across regimes:

$$R_r = \sigma_r^2 \, bb', \qquad r = 1, 2. \tag{4}$$

Throughout, $\lambda_f := \sigma_2^2/\sigma_1^2$ denotes the *own volatility ratio* of the omitted variable, and

$$a_i := (b'q_i)^2 \geq 0$$

its (squared, Σ_1-normalised) *alignment* with the i-th structural direction.

4 The bivariate IV recast: the omitted variable wedge

We now present the problem in the canonical bivariate case of Lewis (2022), which recasts identification through heteroskedasticity as an IV problem. Here, the pencil gap has a transparent reading as instrument strength. The omission produces a *wedge* between the true structural relevance and the relevance the econometrician estimates. In fact, the first-stage relevance is a monotone function of $\delta = |\lambda_2 - \lambda_1|$, so a large gap reads as a strong instrument and a vanishing gap as a weak one. The stakes of the present paper are then immediate: if an omitted variable inflates the observed gap, it inflates the apparent instrument strength, and the weak identification pretest returns a false pass. For the sake of coherence to the paper we are extending, this section adopts the same notation.

4.1 Lewis's recast under correct specification

Take the bivariate structural model in the simultaneous-equations (B-model) form

$$\eta_{1t} = \varepsilon_{1t} + H_{12}\varepsilon_{2t}, \qquad \eta_{2t} = H_{21}\varepsilon_{1t} + \varepsilon_{2t}, \tag{5}$$

with H_{12} the parameter of interest and $\mathrm{Cov}(\varepsilon_t \mid r) = \mathrm{diag}(\sigma^2_{\varepsilon_1,r}, \sigma^2_{\varepsilon_2,r})$ across regimes $r \in \{C, P\}$. Shock 1 has constant variance, $\sigma^2_{\varepsilon_1,C} = \sigma^2_{\varepsilon_1,P}$; shock 2 changes, $\sigma^2_{\varepsilon_2,C} \neq \sigma^2_{\varepsilon_2,P}$. Identification through heteroskedasticity (Rigobon, 2003; Lewis, 2022, eqs. (6)–(8)) uses the endogenous regressor weighted by the demeaned regime indicator as its own instrument,

$$Z_t = w_t \, \eta_{2t}, \qquad w_t = \tfrac{T}{T_P}\mathbf{1}(t \in P) - \tfrac{T}{T_C}\mathbf{1}(t \in C), \tag{6}$$

with first stage $\eta_{2t} = \Pi Z_t + v_t$ and second stage $\eta_{1t} = H_{12}\eta_{2t} + u_t$. Because $E[w_t] = 0$, the relevance is a regime contrast of second moments,

$$E[Z_t \, \eta_{2t}] = E[w_t \, \eta_{2t}^2] \;\propto\; \Delta\sigma^2_{\eta_2} := \mathrm{Var}(\eta_2 \mid P) - \mathrm{Var}(\eta_2 \mid C). \tag{7}$$

Under correct specification $\Delta\sigma^2_{\eta_2} = H_{21}^2\Delta\sigma^2_{\varepsilon_1} + \Delta\sigma^2_{\varepsilon_2} = \Delta\sigma^2_{\varepsilon_2}$ (using $\Delta\sigma^2_{\varepsilon_1} = 0$), so the estimated first-stage relevance equals the structural relevance, $\Pi \propto \Delta\sigma^2_{\varepsilon_2}$, and $\Pi \to 0$ is exactly the weak instrument.

The gap is the instrument's strength. The generalized eigenvalues of the pencil are the variance ratios: $\lambda_1 = 1$ (the constant shock) and $\lambda_2 = \sigma^2_{\varepsilon_2,P}/\sigma^2_{\varepsilon_2,C}$. Hence

$$\delta = \lambda_2 - \lambda_1 = \frac{\Delta\sigma^2_{\varepsilon_2}}{\sigma^2_{\varepsilon_2,C}} \;\propto\; \Delta\sigma^2_{\varepsilon_2} \;\propto\; \Pi. \tag{8}$$

The spectral gap *is* the instrument strength: weak identification ($\delta \to 0$) is the weak instrument ($\Pi \to 0$).

4.2 The wedge under omission

Now add one omitted heteroskedastic factor f_t with loading $b = (b_1, b_2)'$ and $\mathrm{Cov}(f_t \mid r) = \psi_r$:

$$\eta_{1t} = \varepsilon_{1t} + H_{12}\varepsilon_{2t} + b_1 f_t, \qquad \eta_{2t} = H_{21}\varepsilon_{1t} + \varepsilon_{2t} + b_2 f_t. \tag{9}$$

This is exactly the rank-one contamination (4) of our paper, with the factor variances identified as $\sigma_1^2 = \psi_C$, $\sigma_2^2 = \psi_P$, so that $\lambda_f = \psi_P/\psi_C$ and

$$\Delta\psi := \psi_P - \psi_C = \sigma_2^2 - \sigma_1^2.$$

With $\varepsilon_{1t}, \varepsilon_{2t}, f_t$ mutually orthogonal, $\mathrm{Var}(\eta_2 \mid r) = H_{21}^2\sigma_{\varepsilon_1,r}^2 + \sigma_{\varepsilon_2,r}^2 + b_2^2\psi_r$, and taking the regime contrast (again $\Delta\sigma_{\varepsilon_1}^2 = 0$):

Lemma 1 (The first-stage wedge)**.** *The relevance the econometrician estimates is the reduced-form contrast*

$$\Delta\sigma_{\eta_2}^2 = \underbrace{\Delta\sigma_{\varepsilon_2}^2}_{\text{true structural relevance}} + \underbrace{b_2^2\,\Delta\psi}_{\text{wedge from the omission}}, \qquad \textit{i.e.} \qquad \Delta\sigma_{\eta_2}^2 - \Delta\sigma_{\varepsilon_2}^2 = b_2^2\,\Delta\psi. \tag{10}$$

Under correct specification the wedge is zero and $\Delta\sigma_{\eta_2}^2 = \Delta\sigma_{\varepsilon_2}^2$. Under omission with a heteroskedastic factor ($\Delta\psi \neq 0$) and $b_2 \neq 0$, the wedge is nonzero and cannot be separated from the structural term, since the researcher observes only η_t.

4.3 The double blow

The same omission also breaks the exclusion restriction. The second-stage error becomes $u_t = (1 - H_{12}H_{21})\varepsilon_{1t} + (b_1 - H_{12}b_2)f_t$, and since Z_t contains f_t through η_{2t},

$$\mathrm{Cov}(Z_t, u_t) = b_2\,(b_1 - H_{12}b_2)\,\Delta\psi \neq 0, \tag{11}$$

so the instrument is endogenous and $\widehat{H}_{12}$ is inconsistent even under strong identification. This is classic omitted-variable bias on the exclusion restriction – and we record it only to delimit the contribution: unlike the wedge (10), it is not novel.

Both failures, though, are proportional to the *same* $\Delta\psi$. A homoskedastic omitted factor ($\Delta\psi = 0$) neither inflates the F nor breaks exogeneity at first order. When $\Delta\psi \neq 0$, the identical quantity inflates the first-stage relevance (10) and corrupts the instrument (11): the researcher is made *more confident and more biased at once*. The empirically relevant class is precisely an omitted shock that is itself heteroskedastic across the identification regimes – plausible in event-day designs, where the high-volatility window inflates many sources together. The headline is then that weak identification diagnostics are not robust to such omissions: they inflate its apparent relevance, masking the weak identification the test is meant to signal.

5 An interpretable first-order analysis

Rather than tracking the perturbed eigenvalues directly, which is more demanding, we now exploit the variational characterization of generalised eigenvalues. Since Σ_1, Σ_2 are SPD, the computation of the i-th generalised eigenvalue can be recast as an optimisation problem exploiting the Courant–Fischer minimax principle (Golub and Van Loan, 2013):

$$\lambda_i = \min_{\dim(S)=i} \max_{x \in S, x \neq 0} \mathcal{R}(x), \qquad \mathcal{R}(x) := \frac{x'\Sigma_2 x}{x'\Sigma_1 x}. \tag{12}$$

In essence, for any direction x, $x'\Sigma_r x$ is the variance of the linear combination $x'y_t$ under regime r; hence $\mathcal{R}(x)$ is the *relative change in variance* along x when moving from regime-1 to regime-2,

and the generalized eigenvalues are its extremal values.
Under omitted variables the quotient becomes $\widehat{\mathcal{R}}(x) = x'\widehat{\Sigma}_2 x / x'\widehat{\Sigma}_1 x$. Writing

$$v_2 = x'\Sigma_2 x, \quad v_1 = x'\Sigma_1 x, \quad r_2 = x'R_2 x, \quad r_1 = x'R_1 x,$$

where v_1, v_2 are the variances generated by the *observed* variables and r_1, r_2 the additional variance contributed by the omitted variable, we have $\mathcal{R}(x) = v_2/v_1$ and $\widehat{\mathcal{R}}(x) = (v_2+r_2)/(v_1+r_1)$. Then

$$\widehat{\mathcal{R}}(x) - \mathcal{R}(x) = \frac{v_2 + r_2}{v_1 + r_1} - \frac{v_2}{v_1} = \frac{v_1\, r_2 - v_2\, r_1}{v_1(v_1 + r_1)}. \tag{13}$$

This yields the first structural fact. The sign is governed *exclusively* by $v_1\, r_2 - v_2\, r_1$: the extra variance injected into the numerator (regime-2) competes with that injected into the denominator (regime-1). Positive semidefiniteness ($R_1, R_2 \succeq 0 \Rightarrow r_1, r_2 \geq 0$) does *not* fix this sign, so omitted variables may actually raise or lower the quotient. Any systematic result on the gap must therefore exploit additional hypotheses on structure.

5.1 Rank-one structure and the key decomposition

In the context of (4), $r_r(x) = \sigma_r^2 (b'x)^2$ [2], and simplifying terms in (13) renders it more interesting.

Proposition 1 (*Geometry & heteroskedasticity*)**.** *Under* (4)*,*

$$\widehat{\mathcal{R}}(x) - \mathcal{R}(x) = (b'x)^2\, \frac{\sigma_2^2 - \mathcal{R}(x)\, \sigma_1^2}{v_1 + r_1} = (b'x)^2\, \sigma_1^2\, \frac{\lambda_f - \mathcal{R}(x)}{v_1 + \sigma_1^2 (b'x)^2}. \tag{14}$$

Three components appear, and the first two are *completely* separated:

(i) the purely geometric factor $(b'x)^2$, the alignment of the omitted loading with x;

(ii) the heteroskedasticity factor $\sigma_2^2 - \mathcal{R}(x)\sigma_1^2 = \sigma_1^2(\lambda_f - \mathcal{R}(x))$, which *carries the sign*;

(iii) a scale factor $(\, v_1 + \sigma_1^2 (b'x)^2\,)^{-1} > 0$, never altering the sign being it always strictly positive.

What this formula tells us is that even a highly heteroskedastic omitted variable has essentially no effect if it is nearly orthogonal to the relevant directions ($b'x \approx 0$); conversely, a moderately heteroskedastic one can matter a great deal if strongly aligned. Evaluated at $x = q_i$ (so $\mathcal{R}(q_i) = \lambda_i$, $v_1 = 1$, $a_i := (b'q_i)^2$), (14) reads

$$\widehat{\mathcal{R}}(q_i) - \mathcal{R}(q_i) = \frac{a_i\, \sigma_1^2 (\lambda_f - \lambda_i)}{1 + \sigma_1^2 a_i}, \tag{15}$$

and reveals an "attractor" picture: the omitted factor sits at λ_f and pulls λ_i towards it, with force a_i and in the direction of the sign of $\lambda_f - \lambda_i$.

Remark 1. Equation (15) is exact but evaluated at the *fixed* true direction q_i; it is *not* the eigenvalue change $\widehat{\lambda}_i - \lambda_i$, because the eigenvector itself moves, $\widehat{q}_i \neq q_i$. It is nevertheless the correct first-order object (Section 5.2) and the interpretable one, but the exact eigenvalue change requires a secular equation (Section 6).

[2] As $r_r(x) = x'R_r x = x'\sigma_r^2 bb'x = \sigma_r^2 x'bb'x = \sigma_r^2 (b'x)^2$.

5.2 Envelope theorem: first-order shift and gap

Consider now the perturbation embedded in a path $\Sigma_r(t) = \Sigma_r + tR_r$, $t \in [0,1]$, with $\Sigma_r(0)$ the true DGP and $\Sigma_r(1)$ the misspecified model. Let $(\lambda_i(t), q_i(t))$ be the corresponding eigenpairs. Write the path-dependent quotient $\mathcal{R}_t(x) = x'\Sigma_2(t)x / x'\Sigma_1(t)x$. Its gradient in x is, by the quotient rule and $\nabla_x(x'Ax) = 2Ax$ for symmetric A,

$$\nabla_x \mathcal{R}_t(x) = \frac{2}{x'\Sigma_1(t)x}\Big(\Sigma_2(t)x - \mathcal{R}_t(x)\,\Sigma_1(t)x\Big). \tag{16}$$

Evaluated at an eigenvector this gradient vanishes: $\nabla_x \mathcal{R}_t(q_i)$ is the generalised eigenvalue equation $\Sigma_2(t)q_i = \lambda_i(t)\Sigma_1(t)q_i$. Because the Rayleigh quotient is *stationary* at eigenvectors – they are the candidate solutions of the optimisation problem – the envelope theorem (see e.g. Bottazzi (2023)) gives the eigenvalue derivative as the explicit derivative with respect to t of $\mathcal{R}_t$ at the fixed point $x = q_i(t)$, which allows us not to consider the shift in the eigenvector:

Proposition 2 (*Gap derivative via envelope*)**.** *Along the path, with $q_i(t)'\Sigma_1(t)q_i(t) = 1$ for the usual normalisation,*

$$\lambda_i'(t) = \big(b'q_i(t)\big)^2\big(\sigma_2^2 - \lambda_i(t)\sigma_1^2\big) = a_i(t)\,\sigma_1^2\big(\lambda_f - \lambda_i(t)\big). \tag{17}$$

Proof. Since $\mathcal{R}_t(q_i(t)) = \lambda_i(t)$, differentiate $\lambda_i(t) = \mathcal{R}_t(q_i(t))$ by the chain rule:

$$\lambda_i'(t) = \frac{\partial \mathcal{R}_t}{\partial t}\Big|_{x=q_i} + \nabla_q \mathcal{R}_t(q_i(t))\frac{\partial q_i(t)}{\partial t}$$

The second term vanishes: by (16), $\nabla_q \mathcal{R}_t(q_i) = 0$ at the eigenvector. Hence the first-order movement of q_i does not contribute to the eigenvalue. There remains the other term, computed at fixed $x = q_i$ using the quotient rule:

$$\lambda_i'(t) = q_i' R_2 q_i - \lambda_i(t)\, q_i' R_1 q_i,$$

remembering the normalisation $q_i'\Sigma_1(t)q_i = 1$. The rank-one structure $R_r = \sigma_r^2 bb'$ gives $q_i' R_r q_i = \sigma_r^2 (b'q_i)^2$, whence $\lambda_i'(t) = (b'q_i)^2(\sigma_2^2 - \lambda_i\sigma_1^2) = a_i\sigma_1^2(\lambda_f - \lambda_i)$. □

Hence the first-order gap derivative for the binding pair (λ_1, λ_2) – that is, the pair at which the gap reaches its minimum – is

$$\delta'(t) = \lambda_2'(t) - \lambda_1'(t) = \sigma_1^2\Big[a_2(t)\big(\lambda_f - \lambda_2(t)\big) - a_1(t)\big(\lambda_f - \lambda_1(t)\big)\Big]. \tag{18}$$

Its value at $t = 0$, $\delta'(0)$, gives the direction in which the gap first moves under the perturbation.

Why "perturbation always inflates" fails. Equation (18) makes the obstruction to a generalised inflation claim explicit. Suppose $\lambda_f > \lambda_2$ (attractor above both eigenvalues), the only configuration in which both members are pulled upward. Then both $\lambda_f - \lambda_2$ and $\lambda_f - \lambda_1$ are positive, but $\lambda_f - \lambda_2 < \lambda_f - \lambda_1$: the upper member, being closer to the attractor, experiences a smaller attraction force. Gap inflation $\delta'(0) > 0$ thus requires the alignment to concentrate on the upper member enough to overcome its distance handicap, $a_2(\lambda_f - \lambda_2) > a_1(\lambda_f - \lambda_1)$. The geometric factor a_i and the distance factor $\lambda_f - \lambda_i$ pull in opposite directions, so no assumption on the a_i alone delivers $\delta'(0) > 0$. If instead $\lambda_f \in (\lambda_1, \lambda_2)$, then $\lambda_f - \lambda_2 < 0 < \lambda_f - \lambda_1$ and the gap strictly narrows regardless of alignment. This trade-off is genuine; it is the reason why a universal monotonicity theorem does not hold, and it survives to the exact analysis that now follows – but, as we will show, it is confined to a strictly non-degenerate, strong identification regime.

6 Being rigorous: exact secular equation and the gap

Because (Σ_1, Σ_2) is a definite pair, there is a nonsingular Q with

$$Q'\Sigma_1 Q = I, \qquad Q'\Sigma_2 Q = \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n). \tag{19}$$

Writing $d := Q'b$ (so $d_i = q_i'b = b'q_i$, hence $d_i^2 = a_i$), the perturbed pencil $(\widehat{\Sigma}_2, \widehat{\Sigma}_1)$ becomes, in the new basis,

$$\big(\Lambda + \sigma_2^2\, dd',\ I + \sigma_1^2\, dd'\big). \tag{20}$$

The whole problem is now a diagonal pencil updated by a rank-one term *along the same direction* d on both sides. Importantly, no eigenvector of the perturbed pencil is needed here.

6.1 The exact secular equation

The perturbed generalised eigenvalues solve $\det\big((\Lambda - \widehat{\lambda} I) + (\sigma_2^2 - \widehat{\lambda}\sigma_1^2)\, dd'\big) = 0$ [3]. Since $\Lambda - \widehat{\lambda} I$ is diagonal and dd' is rank one, the matrix determinant lemma (Brookes, 2020) yields:

Proposition 3 (Secular equation)**.** *Set $\mu(\hat{\lambda}) := \sigma_2^2 - \hat{\lambda}\,\sigma_1^2$. Then, for every $\hat{\lambda} \notin \{\lambda_1, \ldots, \lambda_n\}$, the scalar $\hat{\lambda}$ is a generalised eigenvalue of the pencil if and only if*

$$1 + \mu(\hat{\lambda}) \sum_{i=1}^{n} \frac{a_i}{\lambda_i - \hat{\lambda}} = 0. \tag{21}$$

Proof. Start from $\det\big((\Lambda - \widehat{\lambda} I) + (\sigma_2^2 - \widehat{\lambda}\sigma_1^2)\, dd'\big) = 0$, and call $D := (\Lambda - \widehat{\lambda} I)$. Fix $\hat{\lambda} \notin \{\lambda_i\}$, so that $D = \mathrm{diag}(\lambda_i - \hat{\lambda})$ is invertible. By the matrix determinant lemma (with general expression $\det(D + uv') = \det(D)\,(1 + v'D^{-1}u)$, applied here with $u = \mu(\hat{\lambda})\, d$ and $v = d$),

$$\det\big(D + \mu(\hat{\lambda})\, dd'\big) = \det(D)\,\Big(1 + \mu(\hat{\lambda})\, d'D^{-1}d\Big).$$

Since D is diagonal, $\det(D) = \prod_i (\lambda_i - \hat{\lambda}) \neq 0$ and, because D^{-1} scales each coordinate by $(\lambda_i - \hat{\lambda})^{-1}$,

$$d'D^{-1}d = \sum_i d_i\, \frac{d_i}{\lambda_i - \hat{\lambda}} = \sum_i \frac{d_i^2}{\lambda_i - \hat{\lambda}} = \sum_i \frac{a_i}{\lambda_i - \hat{\lambda}}.$$

As $\det(D) \neq 0$, vanishing of the determinant is equivalent to vanishing of the second factor, which is (21). □

Remark 2. The eigenvalues for which $\hat{\lambda} = \lambda_i$ are accounted for by the $\det(D)$ expression in the matrix determinant lemma. When a single eigenvalue is perturbed, the other $n - 1$ eigenvalues are the roots of $\prod_i^{n-1}(\lambda_i - \hat{\lambda}) = 0$.

6.2 Closed-form bivariate gap

We now consider the case $n = 2$. The proof of the result that follows, which is algebraically straightforward yet computationally cumbersome, can be found in appendix A of this paper.

[3] To see this, recall we are searching for $\det\big(\widehat{\Sigma}_2 - \widehat{\lambda}\widehat{\Sigma}_1\big) = 0$ from the perturbed GEP. Substituting $\det\big((\Sigma_2 + \sigma_2^2 bb') - \widehat{\lambda}(\Sigma_1 + \sigma_1^2 bb')\big) = 0$ and rearranging yields $\det\big((\Sigma_2 - \widehat{\lambda}\Sigma_1) + (\sigma_2^2 - \widehat{\lambda}\sigma_1^2)bb'\big)$=0. Multiplying left and right both sides by Q' and Q respectively, the determinant is unchanged while the matrices are diagonalised giving the exact form found in the text. We write it in this form because then there exists a closed expression for the determinant of a diagonal matrix plus a rank-one perturbation: the matrix determinant lemma.

Theorem 1 (*Exact spectral gap decomposition*)**.** *Define $S := a_2 + a_1$, and the pull exerted by the omitted factor on eigenvalue i as $\pi_i := a_i(\lambda_f - \lambda_i)$. The perturbation is entirely governed by the differential pull $\pi_2 - \pi_1$ and the total pull $\pi_2 + \pi_1$ so that*

$$\widehat{\delta}^2 - \delta^2 = \frac{2\sigma_1^2\delta\,(\pi_2 - \pi_1) \;+\; (\sigma_1^2)^2\big[(\pi_1 + \pi_2) - 2a_1\delta\big]\big[(\pi_1 + \pi_2) + 2a_2\delta\big]}{(1 + \sigma_1^2 S)^2}. \tag{22}$$

In particular $\widehat{\delta} > \delta$ if and only if the numerator of (22) is positive.

The perturbation therefore cleanly separates into a first-order component driven exclusively by the differential pull and a second-order correction determined by the total pull.

6.3 Consistency with the envelope formula

The exact result reproduces Section 5.2 at leading order for small perturbations, which is precisely what allows us to use the Rayleigh/envelope analysis as the interpretable approximation of the rigorous statement.

Lemma 2 (Envelope is the first-order approximation)**.** *As $\sigma_1^2 \to 0$, (22) gives $\widehat{\delta} - \delta = \sigma_1^2\,(\pi_2 - \pi_1) + O((\sigma_1^2)^2) = \delta'(0) + O((\sigma_1^2)^2)$.*

Proof. In the numerator of (22) the first term is $O(\sigma_1^2)$ and the second is $O((\sigma_1^2)^2)$, while the denominator satisfies $(1 + \sigma_1^2 S)^2 \to 1$ for $\sigma_1^2 \to 0$. Hence

$$\widehat{\delta}^2 - \delta^2 = 2\sigma_1^2\,\delta\,(\pi_2 - \pi_1) + O((\sigma_1^2)^2).$$

Factor the left-hand side as $(\widehat{\delta} - \delta)(\widehat{\delta} + \delta)$; since $\widehat{\delta} \to \delta$ as $\sigma_1^2 \to 0$, one has $\widehat{\delta} + \delta = 2\delta + O(\sigma_1^2)$, so that

$$\widehat{\delta} - \delta = \frac{\widehat{\delta}^2 - \delta^2}{\widehat{\delta} + \delta} = \frac{2\sigma_1^2\,\delta\,(\pi_2 - \pi_1)}{2\delta} + O((\sigma_1^2)^2) = \sigma_1^2\,(\pi_2 - \pi_1) + O((\sigma_1^2)^2).$$

Finally $\sigma_1^2\,(\pi_2 - \pi_1) = \sigma_1^2\big[a_2(\lambda_f - \lambda_2) - a_1(\lambda_f - \lambda_1)\big] = \delta'(0)$ by (18) at $t = 0$, which proves the claim. □

Thus the trade-off diagnosed via the envelope theorem is the genuine leading behaviour, and the "no universal inflation" conclusion is exact, not a mere artefact of linearization: its sign lives in $\pi_2 - \pi_1$, i.e., in the strong-identification regime where δ is bounded away from zero.

6.4 The theorem: lifting at degeneracy

The regime we are interested here is the opposite one: weak identification, when $\delta \to 0$. Here the $O(\delta)$ term of (22) vanishes and only the quadratic σ_1^2 term survives.

Theorem 2 (***Degeneracy lifting***)**.** *Suppose the true pencil is degenerate on the binding pair, $\lambda_1 = \lambda_2 = \lambda$ (so $\delta = 0$: exact nonidentification). Then the observed gap is*

$$\widehat{\delta}\big|_{\delta=0} = \frac{\sigma_1^2\,(a_1 + a_2)\,|\lambda_f - \lambda|}{1 + \sigma_1^2(a_1 + a_2)} \;>\; 0 \tag{23}$$

whenever (i) the omitted variable is heteroskedastic relative to the shocks, $\lambda_f \neq \lambda$, and (ii) it is not orthogonal to the degenerate subspace, $a_1 + a_2 > 0$.

Proof. Set $\delta = 0$ in (22): every term carrying a factor δ vanishes, and the total pull is $\pi_1 + \pi_2 = (a_1 + a_2)(\lambda_f - \lambda) = S(\lambda_f - \lambda)$, so the numerator reduces to $(\sigma_1^2)^2(\pi_1 + \pi_2)^2 = (\sigma_1^2)^2 S^2(\lambda_f - \lambda)^2$. Hence $\widehat{\delta}^2 = (\sigma_1^2)^2 S^2(\lambda_f - \lambda)^2/(1 + \sigma_1^2 S)^2$, and the positive square root is the claim. □

The result stated above is strictly linked to the wedge discussed in Section 4.

The F-form of the lifting theorem. By (8), the true gap tracks $\Delta\sigma^2_{\varepsilon_2}$ while the observed first stage tracks $\Delta\sigma^2_{\eta_2}$; the wedge $b_2^2\Delta\psi$ is therefore the spurious strength of Theorem 2 read on the first-stage F. In the weak identification regime the consequence is sharp: when the true relevance vanishes, $\Delta\sigma^2_{\varepsilon_2} \approx 0$ (equivalently $\delta \approx 0$), a heteroskedastic omission still delivers $\Delta\sigma^2_{\eta_2} = b_2^2\Delta\psi \neq 0$, so the observed first stage is bounded away from zero, the F is inflated, and the weak instrument pretest passes. Standard inference then proceeds on an instrument that the diagnostic has wrongly certified – a false green light, with the attendant size distortion.

Remark 3 (Two representations of the same wedge). The lifting theorem (Th. 2) and Lemma 1 are the spectral and first-stage-F representations of the same object. Both vanish iff $\Delta\psi = 0$: a *homoskedastic* omission is differenced away by the regime demeaning in (6) and inflates neither the gap nor the F at first order. Both are proportional to $\Delta\psi$, and both certify apparent strength exactly where the DGP is weak. Consistently with the paper's attractor framework, relative to $\lambda_1 = 1$ the induced shift is $\propto (\lambda_f - 1) \propto \Delta\psi$: the dangerous class is an omitted shock that is itself heteroskedastic across the identification windows.

We now provide two simple results which extend the scope of Th. 2 to weak identification settings and higher dimensional settings respectively.

Corollary 1 (Robustness in the weak identification regime)**.** *Write* $L := \dfrac{\sigma_1^2(a_1+a_2)|\lambda_f - \lambda|}{1+\sigma_1^2(a_1+a_2)} > 0$ *for the gap of (23). For every* ε *with* $0 < \varepsilon < L$ *there exists* $\delta_0 > 0$ *such that every true gap* $\delta < \delta_0$ *satisfies*

$$\widehat{\delta} \;\geq\; L - \varepsilon \;>\; \delta.$$

Hence in the weak identification regime $\widehat{\delta} > \delta$ *holds robustly.*

Proof. The observed gap (23) is a continuous function of the true gap δ. By Theorem 2, $\widehat{\delta} = L$ at $\delta = 0$. Continuity therefore gives a $\delta_1 > 0$ such that $\widehat{\delta} \geq L - \varepsilon$ whenever $\delta < \delta_1$. Setting $\delta_0 := \min\{\delta_1, L - \varepsilon\} > 0$, any $\delta < \delta_0$ satisfies both $\widehat{\delta} \geq L - \varepsilon$ and $L - \varepsilon > \delta$, whence the claim. □

Corollary 2. *Theorem 2 extends to the* n*-dimensional setting, for the eigenvalue that is perturbed by the rank-one contamination:*

$$\widehat{\delta}_{part}\big|_{\delta=0} = \frac{\sigma_1^2\, S\, |\lambda_f - \lambda|}{1+\sigma_1^2 S} \;>\; 0 \tag{24}$$

where $S = \sum_{i=1}^n a_i$*. All the other gaps are still degenerate. In this sense, misspecification in the* n*-dimensional setting mimics partial identification rather than full.*

Proof. In the degenerate n-dimensional regime $\lambda_i = \lambda \;\forall i = 1, ..., n$, so $D = (\lambda - \hat{\lambda})I$ and the matrix determinant lemma gives

$$\det(D + \mu dd') = (\lambda - \hat{\lambda})^n \Big(1 + \mu \tfrac{S}{\lambda - \hat{\lambda}}\Big) = (\lambda - \hat{\lambda})^{n-1}\big[(\lambda - \hat{\lambda}) + \mu S\big] = 0,$$

with $\mu = \sigma_2^2 - \hat{\lambda}\sigma_1^2$ and $S = \sum_i a_i$. The factor $(\lambda - \hat{\lambda})^{n-1}$ leaves $n-1$ eigenvalues at λ, while the bracket gives the single lifted root $\hat{\lambda} = (\lambda + \sigma_2^2 S)/(1 + \sigma_1^2 S)$, whence $\widehat{\delta}_{\text{part}} = \hat{\lambda} - \lambda = \sigma_1^2 S(\lambda_f - \lambda)/(1 + \sigma_1^2 S)$. □

The trade-off of Section 5.2 and the clean lifting of Theorem 2 are two faces of the same exact formula. Where identification is strong (δ large) the omitted variable may enlarge or shrink the gap, and which way it goes is impossible to determine uniformly. Where identification is weak (δ small) – **exactly where diagnostics are meant to raise the alarm** – a heteroskedastic, non-orthogonal omission artificially creates a strictly positive spectral gap out of a degeneracy. This gap can potentially be very large with obvious disruptive repercussions on the quality of inference.

What this establishes for Econometrics

We separate two claims that are easy to conflate.

Remark 4 (The diagnostic reads the misspecified pencil). Weak-identification statistics are functions of the observed covariances $\widehat{\Sigma}_r$, hence a monotone function of $\widehat{\delta}$, the spectral separation of the *misspecified* pencil. They therefore measure the strength of identification of the estimated model, not necessarily that of the structural DGP.

This is a logical consequence of applying the test to the estimated covariances and needs no further result.

Proposition 4 (Masking of weak identification)**.** *In the weak identification regime (δ small, up to exact degeneracy $\delta = 0$), a heteroskedastic omitted variable ($\lambda_f \neq \lambda$) with nonzero alignment to the degenerate subspace inflates the observed gap, $\widehat{\delta} > \delta$, by Theorem 2 and Corollary 1. Consequently, the diagnostic reports strong identification precisely where the DGP is weakly identified: misspecification masks the weak identification of the structural DGP.*

This is the substantive statement of this paper, and it is backed by an exact theorem – conditional on the weak identification regime. The latter restriction is not a weakness but the *economically relevant domain*: the pretests exist to flag weak identification, and that is exactly where the lifting operates. Diagnostic pre-tests are not resilient to this specific kind of misspecification. Under omission the eigenvectors move, so the estimated structural directions $\widehat{b}_i$ are *different objects* from the true b_i. The test does not malfunction: it faithfully reports the separation of the misspecified pencil, whose shocks happen not to be the ones of interest. "Masking weak identification" is therefore a precise expression only once one anchors its scope on wanting the DGP's structural shocks – which is of course the standing assumption here.

7 Monte Carlo evidence

The results of the previous sections are population statements. This section asks whether they survive estimation noise and, more crucially, whether they are large enough to mislead a formal test of identification. For both questions the answer is positive, and we also find – a point we had anticipated in Corollary 2 – that a single omitted variable can make a system in which *nothing* is identified look exactly like one that is *partially identified*.

7.1 Design

We do not build a simulation from scratch. Instead, we take the replication code of Lütkepohl et al. (2021) and change **one line** of it. Their DGP (1b) sets $n = 2$, $\Sigma_1 = I_2$ and $\Sigma_2 = \mathrm{diag}(2, 2)$, and their DGP (8b) sets $n = 3$, $\Sigma_1 = I_3$ and $\Sigma_2 = \mathrm{diag}(2, 2, 2)$; in both cases the volatility ratios are *exactly* equal, so the true spectral gap is zero, nothing is identified, and their null hypotheses of eigenvalue equality are true. These are the designs they use to assess the size of their test, and they are precisely the degeneracy configuration of Theorem 2. This allows us to borrow their code for our simulations.

To these true DGPs, then, we add one omitted variable: a scalar factor f_t with loading b and variance change so that $\widehat{\Sigma}_r = \Sigma_r + \sigma_r^2 bb'$ exactly as in (4). The econometrician still estimates the n-variable VAR on y_t alone. Everything else – the iterated GLS estimation of the VAR coefficients, the simultaneous diagonalisation, the kurtosis estimators, the Wald statistic Q_r and the sequential testing logic – is their code, without further modifications. Because $\Sigma_1 = I$ in these designs the eigenvectors are the coordinate axes, so $a_i = b_i^2$ and the lift predicted by Theorem 2 is available in closed form via (23). As a check on the port, we first reproduced their published Table 1 and Table 4; the replicated rejection frequencies agree with theirs within Monte Carlo error.

7.2 The bivariate case

Table 1 reports rejection frequencies of $H_0 : \lambda_1 = \lambda_2$ in DGP (1b). Since the null is true, every rejection is a false declaration of identification. Three features stand out.

First, the control row is decisive for the interpretation of the mechanism. When the omitted factor has the same relative volatility as the shocks, $\lambda_f = \lambda$, the gap is not lifted, and the test retains its nominal size. The mere presence of an omitted variable is therefore harmless; what matters is its heteroskedasticity *relative* to the shocks, which is condition (i) of Theorem 2.

Second, rejection frequencies increase monotonically in the predicted lift, from essentially nominal size to certainty. The two columns reporting the lift are computed independently – one from the closed-form expression of Theorem 2, the other as the exact generalized eigenvalue gap of the contaminated pencil – and they coincide exactly.

Third, and diagnostically the most telling, the distortion *worsens* as T grows: for a lift of 1.33 the rejection frequency rises from 0.21 at $T = 100$ to 0.73 at $T = 500$. **This is the signature of a population phenomenon rather than a finite-sample artefact**. More data sharpen the contaminated gap just as they would sharpen a genuine one, so no amount of data protects against the masking; consistency works against the researcher here.

Table 1: Omitted-variable gap lifting and the identification test (bivariate case)

				Rejection frequency			
Scenario	S	lift	$\widehat{\delta}$ (pop.)	$T = 100$	$T = 250$	$T = 500$	median $\widehat{gap}$
baseline: no omission	0.00	0.000	0.000	0.063	0.055	0.058	0.424
control: $\lambda_f = \lambda$	2.00	0.000	0.000	0.066	0.057	**0.050**	0.421
$\lambda_f = 3$	2.00	0.667	0.667	0.106	0.166	**0.284**	0.794
$\lambda_f = 4$	2.00	1.333	1.333	0.212	0.428	**0.727**	1.419
$\lambda_f = 8$	2.00	4.000	4.000	0.689	0.978	**1.000**	4.083
$\lambda_f = 0.5$	2.00	1.000	1.000	0.338	0.689	0.943	1.038
$\lambda_f = 4$, $b = (1, 0)$	1.00	1.000	1.000	0.149	0.291	0.510	1.090
$\lambda_f = 4$, $b = (.5, .5)$	0.50	0.667	0.667	0.107	0.165	0.280	0.791
$\lambda_f = 4$, $\sigma_1^2 = 0.25$	0.50	0.222	0.222	0.064	0.066	0.077	0.486

Note DGP (1b) of Lütkepohl et al. (2021): $n = 2$, $\lambda_1 = \lambda_2 = 2$, $\tau = 0.5$, Gaussian errors, VAR(0), known volatility break. The true gap is zero, so the null $H_0 : \lambda_1 = \lambda_2$ of non-identification is *true* everywhere and every rejection is a false claim of identification. "lift" is the population gap predicted by Theorem 2; $\widehat{\delta}$ (pop.) is the exact generalized eigenvalue gap of the contaminated pencil. Nominal level 5%; 10000 replications. Median $\widehat{gap}$ refers to $T = 500$ and is the sample analogue of $\widehat{\delta}$.

7.3 Spurious partial identification

Table 2 turns to $n = 3$ with all three volatility ratios equal, and applies the sequential procedure of Lütkepohl et al. (2021). A key finding here is that the rank of the contamination matters. A single omitted variable perturbs the covariance matrices along one direction only, so of the three degenerate directions it lifts exactly one: the population eigenvalues become $(\lambda', \lambda, \lambda)$ with $\lambda' \neq \lambda$. One pair therefore remains genuinely degenerate no matter how strong the contamination. The geometry is transparent. Consider the one-standard-deviation surface of the second-regime covariance in the whitening metric of regime-1. When the three volatility ratios coincide this surface is a sphere in $\mathbb{R}^3$: all directions are equally variable and no axis is distinguishable, which is exactly the failure of identification. A rank-one perturbation acts along the single direction b; on its orthogonal complement, a two-dimensional subspace, one has $b'q = 0$, so the alignment a_i vanishes and those directions are left untouched. The sphere is thus stretched along one axis only: the two planes spanned by b and an orthogonal direction acquire elliptical sections,

whereas the plane orthogonal to b remains circular. Identification is restored in the former and remains absent in the latter. More generally, the rank of the perturbation bounds the number of degenerate directions it can separate: with k omitted variables at most k of the degenerate directions are lifted, so a rank-deficient contamination can never mimic *full* identification – which is what the last column of Table 2 records.

The sequential procedure reads this configuration in the natural way, and is wrong in a specific manner. It rejects the global null of non-identification and rejects one of the two pairwise nulls, while failing to reject the other: the frequency with which all three nulls are rejected never departs materially from the nominal level. In other words, the omission does not manufacture the appearance of *full* identification – it manufactures the appearance of *partial* identification. Which pair appears identified is governed by the position of the attractor: for $\lambda_f > \lambda$ the lifted eigenvalue moves above the group and H_{02} is rejected, while for $\lambda_f < \lambda$ it moves below and the roles of H_{02} and H_{03} are exchanged.

The last two blocks of Table 2 make the point sharper than any statement about size distortions. With $\lambda_f = 4$, $\sigma_1^2 = 1$ and $b = (1, 0, 0)$ the contaminated pencil has population eigenvalues $(3, 2, 2)$, which are exactly the eigenvalues of their DGP (7b) – a design in which the first shock genuinely is identified and the remaining two genuinely are not. The two rejection patterns are statistically indistinguishable: at $T = 500$ the global null is rejected with frequency 0.509 in both cases, and the first pairwise null with frequency 0.288 against 0.287, differences well below one Monte Carlo standard error. A system in which nothing is identified, contaminated by one omitted variable, is empirically indistinguishable from a partially-identified one.

It is worth being explicit about what this does and does not say about the test. The procedure is not malfunctioning: it is delivering correct inference about the spectral separation of the pencil it is given. That pencil is simply not the one implied by the data-generating process, due to the model's misspecification.

Table 2: Spurious partial identification from a single omitted variable

			Rejection frequency			
Scenario	Population λ	T	H_{01}	$H_{02} \mid \neg H_{01}$	$H_{03} \mid \neg H_{01}$	all three
control: $\lambda_f = \lambda$	(2.00, 2.00, 2.00)	100	0.077	0.017	0.015	0.000
		500	0.055	0.013	0.010	0.000
$\lambda_f = 4$, $b = (1, 1, 1)$	(3.50, 2.00, 2.00)	100	0.239	0.095	0.031	0.001
		500	**0.823**	**0.636**	**0.044**	0.015
$\lambda_f = 8$	(6.50, 2.00, 2.00)	100	0.794	0.567	0.059	0.018
		500	1.000	1.000	0.049	0.049
$\lambda_f = 0.5$	(2.00, 2.00, 0.88)	100	0.429	0.045	0.225	0.004
		500	0.992	0.052	0.970	0.045
$\lambda_f = 4$, $b = (1, 0, 0)$	(3.00, 2.00, 2.00)	100	0.153	0.051	0.026	0.001
		500	**0.509**	**0.288**	**0.040**	0.003
reference: DGP (7b), no omission	(3.00, 2.00, 2.00)	100	0.153	0.045	0.022	0.000
		500	**0.509**	**0.287**	**0.033**	0.002

Note DGP (8b) of Lütkepohl et al. (2021): $n = 3$, $\lambda_1 = \lambda_2 = \lambda_3 = 2$, so *no* shock is identified and all three nulls are true. Sequential procedure with $Q_r(\widehat{\kappa}_1, \widehat{\kappa}_2)$: $H_{01} : \lambda_1 = \lambda_2 = \lambda_3$, $H_{02} : \lambda_1 = \lambda_2$, $H_{03} : \lambda_2 = \lambda_3$, the latter two evaluated only on replications where H_{01} is rejected. The last block is their genuinely partially identified DGP (7b) with no omitted variable: its rejection pattern is statistically indistinguishable from the contaminated one. Nominal level 5%; 5000 replications.

7.4 Relation to Lütkepohl et al. (2021)

The two exercises are complementary rather than competing. Their Monte Carlo establishes that, under correct specification, the test attains its nominal size and has power that increases in the true eigenvalue separation. Our experiment holds their test fixed and perturbs the specification instead, and the failure it produces is located in the size rather than the power of the test: a true null of non-identification is rejected far too often, so the sequential procedure certifies identifying information that the data-generating process does not contain. Nothing in their asymptotic theory is contradicted, because their maintained assumption is that the two-regime decomposition of the observed covariances is correctly specified. The mechanism operates entirely outside that assumption.

The concern is in fact one the authors themselves raise in a different form. In over-parameterised specifications they find the test to be oversized in small samples (p. 12) and observe that this may lead an investigator to overstate the identifying information present in the second moments. Our results can be read as the false-DGP counterpart of that warning: the same false conclusion arises from an incomplete variable set, does not vanish as T grows, and cannot be diagnosed from the estimated covariances alone, since the statistic is a function of exactly those covariances.

8 Conclusions

This paper shows that an omitted variable can generate spurious identifying power in structural VAR models employing the Changes in Volatility identification approach. Even a single omitted factor can affect the generalised eigenvalues of the covariance pencil and create an artificial gap when the underlying system is not identified. Consequently, standard tests based on spectral separation may reject a true null of non-identification, with the distortion becoming more pronounced as the sample size increases. The analysis also shows that the magnitude of this deception depends on both the heteroskedasticity of the omitted factor and its alignment with the relevant eigenspaces. These results suggest that spectral separation should not be interpreted solely as evidence of structural identification, at least without considering the possibility of misspecification. More importantly, the peril our mechanism identifies is not the standard asymptotic failure of weak identification, but something quieter and worse: misspecification suppresses the very signal that would prompt the search for an alternative identification method. By inflating the observed gap, a heteroskedastic omitted variable returns a false pass; the researcher, seeing an apparently strong instrument, has no reason to turn to weak-ID robust inference or to a different identifying strategy. The failure that weak identification would have announced is silenced, and standard inference proceeds on a foundation the DGP does not support. And of course, the omission hits twice: the same factor that silences the alarm also biases the estimates, as noted in Section 4.3.

In our paper we focused on a purely contemporaneous, rank-one omission. This choice was made not only for the sake of simplicity and clarity, but also because, intuitively, adding other linearly independent terms such as other variables or their lagged dynamics would make the case for even worse masking. We discuss this only informally, in Appendix B. Thus, two directions appear particularly relevant for future research. First, extending the present analysis to higher-rank contaminations would clarify how multiple omitted channels can generate spurious separation across several eigenspaces, potentially leading to spurious full rather than partial identification. Second, allowing the omitted variable to have its own dynamics would make it possible to study the trade-off between a potentially richer contamination structure and the attenuation of misspecification when part of the omitted dynamics can be predicted from the included variables (e.g., possibly increasing lag length). Both extensions would help assess the extent to which the masking mechanism survives in more general forms of misspecification.

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A Analytical appendix

Throughout, $s := \sigma_1^2$, $\lambda_f := \sigma_2^2/\sigma_1^2$, $\delta := \lambda_2 - \lambda_1 > 0$, and $a_i := (b'q_i)^2$ is the alignment of the omitted loading with structural direction i. The single object carrying interpretation is the *pull* exerted by the attractor λ_f on eigenvalue i,

$$\pi_i := a_i\,(\lambda_f - \lambda_i) \tag{25}$$

which is exactly the envelope drift of Proposition 2, $\lambda_i'(0) = \sigma_1^2\,\pi_i$. Two combinations of the pulls above are needed below: the *differential pull* $\pi_2 - \pi_1$ and the *total pull* $\pi_1 + \pi_2$. We also make the abbreviations $S := a_1 + a_2$ (the total alignment) and $\kappa := a_1\lambda_2 + a_2\lambda_1$.

Clearing out the denominator of (21), the perturbed eigenvalues solve the quadratic $\alpha\widehat{\lambda}^2 + \beta\widehat{\lambda} + \gamma = 0$ with

$$\alpha = 1 + sS, \qquad \beta = -\big[(\lambda_1 + \lambda_2) + s(\lambda_f S + \kappa)\big], \qquad \gamma = \lambda_1\lambda_2 + s\lambda_f\kappa. \tag{26}$$

Computing $\widehat{\lambda}_+$ and $\widehat{\lambda}_-$ and squaring their difference we obtain

$$\widehat{\delta}^{\,2} = \frac{\beta^2 - 4\alpha\gamma}{\alpha^2}. \tag{27}$$

Since $\alpha > 0$, the whole proof is the evaluation of $\beta^2 - 4\alpha\gamma$.

Two elementary contractions

Lemma 3. *With $\delta = \lambda_2 - \lambda_1$,*

$$(\lambda_1 + \lambda_2)\,S - 2\kappa = (a_2 - a_1)\,\delta, \qquad (\lambda_1 + \lambda_2)\,\kappa - 2\,\lambda_1\lambda_2\,S = (a_1\lambda_2 - a_2\lambda_1)\,\delta.$$

Proof. Direct expansion. For the first,

$$(\lambda_1 + \lambda_2)(a_1 + a_2) - 2(a_1\lambda_2 + a_2\lambda_1) = a_1(\lambda_1 - \lambda_2) + a_2(\lambda_2 - \lambda_1) = (a_2 - a_1)(\lambda_2 - \lambda_1).$$

For the second,

$$(\lambda_1+\lambda_2)(a_1\lambda_2+a_2\lambda_1)-2(a_1+a_2)\lambda_1\lambda_2 = a_1\lambda_2(\lambda_2-\lambda_1)+a_2\lambda_1(\lambda_1-\lambda_2) = (a_1\lambda_2-a_2\lambda_1)(\lambda_2-\lambda_1).$$

□

The discriminant

$$\beta^2 - 4\alpha\gamma = \delta^2 + 2s\delta\big[(\pi_2 - \pi_1) + S\delta\big] + s^2\big[(\pi_1 + \pi_2) + (a_2 - a_1)\delta\big]^2. \tag{28}$$

Proof. From (26),

$$\beta^2 = (\lambda_1+\lambda_2)^2+2s(\lambda_1+\lambda_2)(\lambda_f S+\kappa)+s^2(\lambda_f S+\kappa)^2, \quad 4\alpha\gamma = 4\lambda_1\lambda_2+4s(\lambda_f\kappa+\lambda_1\lambda_2 S)+4s^2 S\lambda_f\kappa.$$

Subtracting and collecting by powers of s,

$$\beta^2-4\alpha\gamma = \underbrace{(\lambda_1 + \lambda_2)^2 - 4\lambda_1\lambda_2}_{s^0} + 2s\,\underbrace{\big[(\lambda_1 + \lambda_2)(\lambda_f S + \kappa) - 2\lambda_f\kappa - 2\lambda_1\lambda_2 S\big]}_{s^1} + s^2\,\underbrace{\big[(\lambda_f S + \kappa)^2 - 4S\lambda_f\kappa\big]}_{s^2}.$$

— *Order s^0.* $(\lambda_1 + \lambda_2)^2 - 4\lambda_1\lambda_2 = (\lambda_2 - \lambda_1)^2 = \delta^2$.

— *Order s^2.* A perfect square, $(\lambda_f S + \kappa)^2 - 4S\lambda_f\kappa = (\lambda_f S - \kappa)^2$, and

$$\lambda_f S - \kappa = \lambda_f(a_1 + a_2) - (a_1\lambda_2 + a_2\lambda_1) = a_1(\lambda_f - \lambda_2) + a_2(\lambda_f - \lambda_1) = (\pi_1 + \pi_2) + (a_2 - a_1)\delta,$$

using $a_1(\lambda_f - \lambda_2) = \pi_1 - a_1\delta$ and $a_2(\lambda_f - \lambda_1) = \pi_2 + a_2\delta$.

— *Order* s^1. Group as $\lambda_f\big[(\lambda_1+\lambda_2)S - 2\kappa\big] + \big[(\lambda_1+\lambda_2)\kappa - 2\lambda_1\lambda_2 S\big]$ and apply Lemma 3:

$$\lambda_f(a_2 - a_1)\delta + (a_1\lambda_2 - a_2\lambda_1)\delta = \delta\big[\lambda_f(a_2 - a_1) + a_1\lambda_2 - a_2\lambda_1\big] = \delta\big[(\pi_2 - \pi_1) + S\delta\big],$$

the last step being $a_2(\lambda_f - \lambda_1) - a_1(\lambda_f - \lambda_2) = (\pi_2 - \pi_1) + S\delta$. Substituting the three coefficients gives (28). □

Proof of Theorem 1

Proof. By (27), $\widehat{\delta}^2 - \delta^2 = (\beta^2 - 4\alpha\gamma - \delta^2\alpha^2)/\alpha^2$ with $\alpha^2 = (1+sS)^2 = 1 + 2sS + s^2S^2$. Using (28), the numerator is

$$\delta^2 + 2s\delta\big[(\pi_2 - \pi_1) + S\delta\big] + s^2\big[(\pi_1 + \pi_2) + (a_2 - a_1)\delta\big]^2 - \delta^2\big(1 + 2sS + s^2S^2\big).$$

The δ^2 terms cancel, and so do the two $2s\delta^2 S$ terms, leaving

$$2s\delta\,(\pi_2 - \pi_1) + s^2\Big\{\big[(\pi_1 + \pi_2) + (a_2 - a_1)\delta\big]^2 - S^2\delta^2\Big\}.$$

The brace is a difference of squares with $S = a_1 + a_2$; since $(a_2 - a_1) - (a_1 + a_2) = -2a_1$ and $(a_2 - a_1) + (a_1 + a_2) = 2a_2$,

$$\big[(\pi_1 + \pi_2) + (a_2 - a_1)\delta\big]^2 - S^2\delta^2 = \big[(\pi_1 + \pi_2) - 2a_1\delta\big]\big[(\pi_1 + \pi_2) + 2a_2\delta\big].$$

Dividing by α^2 gives (22). As $\widehat{\delta}, \delta > 0$ the sign of $\widehat{\delta}^2 - \delta^2$ equals that of $\widehat{\delta} - \delta$, and $\alpha^2 > 0$, so inflation is governed by the numerator. □

B Non-contemporaneous omission

Let the true data-generating process (DGP) be a VAR(p) with $n+1$ variables:

$$y_t = A_1 y_{t-1} + ... + A_p y_{t-p} + B\epsilon_t,$$

where $y_t \in \mathbb{R}^{n+1}$, $A_1, ..., A_p, B \in \mathbb{R}^{(n+1)\times(n+1)}$, and $\epsilon_t \in \mathbb{R}^{n+1}$ is a vector of mutually uncorrelated structural shocks. Now, suppose the researcher observes only $\tilde{y}_t \in \mathbb{R}^n$, the first n components of y_t. Partition the DGP conformably:

$$y_t = \begin{pmatrix} \tilde{y}_t \\ y_{(n+1)t} \end{pmatrix}, \quad A_i = \begin{pmatrix} \tilde{A}_i & a_i \\ * & * \end{pmatrix}, \quad B = \begin{pmatrix} \tilde{B} & b \\ * & * \end{pmatrix},$$

where $\tilde{A}_i \in \mathbb{R}^{n\times n}$, $a_i \in \mathbb{R}^{n\times 1}$ for $i = 1, ..., p$, $\tilde{B} \in \mathbb{R}^{n\times n}$, and $b \in \mathbb{R}^n$ is the impulse vector of the omitted shock $\epsilon_{(n+1)t}$. The true process for the observed variables is

$$\tilde{y}_t = \tilde{A}_1\tilde{y}_{t-1} + ... + \tilde{A}_p\tilde{y}_{t-p} + a_1 y_{(n+1),t-1} + ... + a_p y_{(n+1),t-p} + \tilde{B}\tilde{\epsilon}_t + b\,\epsilon_{(n+1)t},$$

whereas the researcher estimates $\tilde{y}_t = \hat{A}_1\tilde{y}_{t-1} + ... + \hat{A}_p\tilde{y}_{t-p} + \hat{u}_t$. Provided the VAR is stationary, then, the reduced-form residuals converge in probability to

$$\hat{u}_t \xrightarrow{p} \underbrace{\tilde{B}\tilde{\epsilon}_t}_{\tilde{u}_t} + \underbrace{b\,\epsilon_{(n+1)t} + a_1 y_{(n+1),t-1} + ... + a_p y_{(n+1),t-p} + (\tilde{A}_1 - \hat{A}_1)\tilde{y}_{t-1} + ... + (\tilde{A}_p - \hat{A}_p)\tilde{y}_{t-p}}_{\eta_t},$$

where $\tilde{u}_t$ are the clean residuals and η_t is the total contamination, combining the contemporaneous impact of $\epsilon_{(n+1)t}$ and its lagged dynamics. The coefficient matrices $\hat{A}_i$ may well not converge

to their true values $\tilde{A}_i$, adding to the contamination term. Notice, then, how $\hat{u}_t$ violates the hypothesis of white noise residuals due to the serial correlation of the contamination terms. The covariance of $\hat{u}_t$ in regime k converges to

$$\hat{\Sigma}^{(k)} = \tilde{\Sigma}^{(k)} + P_k,$$

where $\tilde{\Sigma}^{(k)} = \tilde{B}\tilde{\Lambda}^{(k)}\tilde{B}^T$ is the uncontaminated $n \times n$ covariance and

$$P_k = \mathbb{E}[\eta_t \eta_t^T \mid k] \tag{29}$$

captures both the contemporaneous variance of the omitted shock and the autocovariance structure generated by the lagged dynamics of $y_{(n+1)t}$. We stress that, generically, $P_1 \neq P_2$ because we allow for changes in the variance of the omitted variable. Notice that $\mathbb{E}[\tilde{u}_t \eta_t^T]$ and $\mathbb{E}[\eta_t \tilde{u}_t^T]$ both equal zero because of the definition of structural shock.

Beyond rank-one. Unlike the rank-one $R_r = \sigma_r^2 bb'$ of the main text, P_k above aggregates the contemporaneous impact of $\epsilon_{(n+1)t}$, the autocovariance generated by its own lagged dynamics $a_1, \ldots, a_p$, and the coefficient bias $(\tilde{A}_i - \hat{A}_i)$ induced by the omission – a sum of several distinct random components rather than a single scaled outer product. Generically, then, P_k is not rank-one, or even rank-bounded by the number of omitted variables: it is a full-rank $n \times n$ matrix.

This has direct consequences for the masking mechanism. By Corollary 2, a rank-one contamination can lift at most one eigenvalue out of a degenerate cluster, so a wholly unidentified system can only ever be masked as *partially* identified – never as fully identified. A full-rank P_k removes this floor: a perturbation of rank r can displace up to r eigenvalues, so a full-rank contamination is, in principle, capable of lifting every degenerate direction simultaneously. We conjecture that, generically, the dynamic omission of Assumption 1 therefore worsens rather than mitigates the masking documented in Section 6, potentially turning a wholly unidentified DGP into one that appears *fully* identified rather than partially. We do not formalise this claim here; a full characterisation of the eigenvalue shift induced by a general SPD perturbation, as opposed to the rank-one case, is left to future work.